A3-PS019-TC210

Personal-Data Discovery and Privacy Evidence Inventory

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Background

Does your organization know exactly where all its personal data lives?

Most companies — even large ones — don't.

Here's what happens in the real world. Over the years, customer data gets scattered across CSV files, spreadsheets, and database tables. A field like a phone number or an Aadhaar number ends up sitting in some unindexed table, and nobody has documented why it's there or how long it's kept.

Now, modern privacy laws — like India's DPDP Act — require organizations to know:

- What personal data they store,
- Why they store it — the purpose, and
- How long they keep it — the retention period.

But when data spreads across systems without documentation, this becomes nearly impossible to track.

Problem Description

The core problems this project addresses are:

- **Data Sprawl**: PII accumulates in disparate locations like CSV files and SQL databases without centralized documentation or oversight.
- **Lack of Context**: Discovered data fields are rarely linked to their declared business purpose or required retention schedule, making it impossible to enforce data governance policies.
- **Inefficient Discovery**: Manual data discovery is time-consuming, error-prone, and fails to scale, leaving large volumes of "dark data" unmanaged.
- **Compliance Risk**: Without a clear inventory, organizations cannot effectively respond to Data Subject Access Requests (DSARs) or demonstrate compliance with privacy regulations.
- **Evidence Gap**: There is a critical need for a technical tool that can produce an evidence-based inventory for human review, clearly distinguishing factual findings from legal interpretation.

Literature & RFC Review

Our design is grounded in India's DPDP framework, informed by IETF privacy RFCs, and validated by recent literature showing that hybrid rule-based + lightweight classification approaches are effective and explainable for PII discovery

1. Data Protection & Privacy Frameworks (Regulatory Context)

- Digital Personal Data Protection (DPDP) Act, 2023 & DPDP Rules, 2025 (MeitY, Government of India)
- MeitY Data Protection Framework
- NIST Privacy Framework 1.0

2. RFCs & Internet Privacy Guidelines

- RFC 6973: Privacy Considerations for Internet Protocols (IETF, 2013)
- RFC 8280 / RFC 9620: Human Rights Protocol and Architecture Considerations (IRTF)

3. PII Detection & Data Discovery Literature

- Comprehensive Review of PII Leakage Detection Using AI Techniques (IEEE, 2025)
- Hybrid Rule-Based NLP and ML Approach for PII Detection (Scientific Reports, 2025)
- Detection of PII Leakage on the Web Using BERT-BiLSTM-Attention (Semantic Scholar)

Proposed Solution : PriVista

PriVista is a web-based personal-data discovery and privacy evidence scanner. It inspects CSV files and a SQL database to identify candidate PII fields, links them to declared purposes and retention schedules, and flags missing metadata for human review.

- **Multi-Source Ingestion:** Accepts CSV uploads and connects to a SQL database (PostgreSQL/MySQL) via a form-based panel.
- **Configurable PII Detection**: A built-in rules.yaml defines regex and keyword patterns across five data classes — Identity, Contact, Financial, Health, and Online Identifiers. Users may override rules without touching code.
- **Metadata Linking:** Cross-references every detected PII field against a user-uploaded metadata.yaml containing declared purposes and retention schedules.
- **Gap Flagging:** Fields detected as PII but not declared in metadata are flagged as evidence gaps.
- **Self-Evaluation:** Against a small manually-labelled test set, PriVista reports per-class precision and recall.
- **Structured Output:** Generates a downloadable JSON/PDF evidence inventory for auditors and privacy teams.
- **Clear Scope Boundary:** The tool produces technical evidence only — it does not make legal compliance decisions or infer data flows.

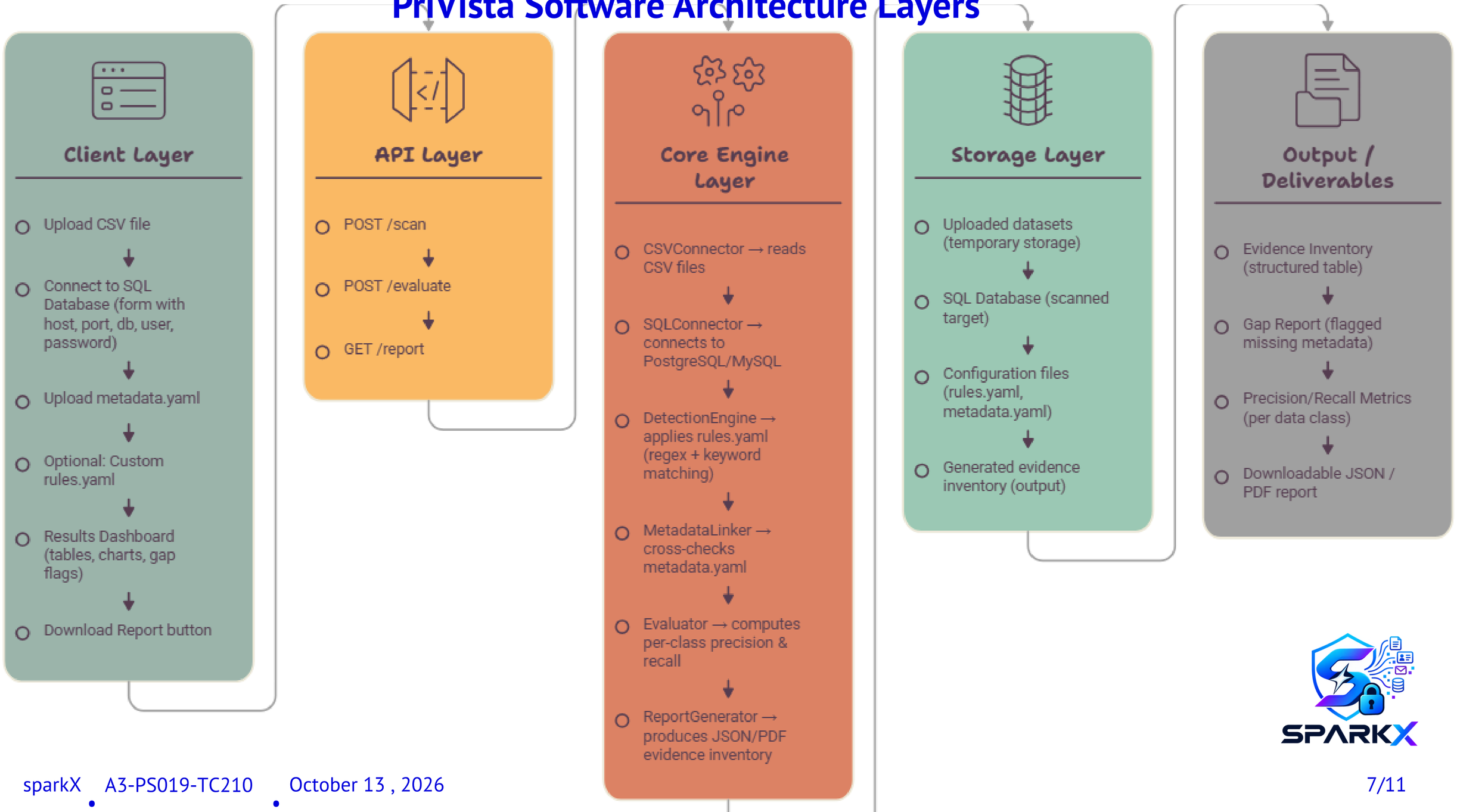
Optimization Proposed

- **PriVista replaces a slow, manual, and inconsistent discovery process with a fast, reproducible, and auditable one.**

Comparison of Manual Process and PriVista Solution

Feature	Speed	Accuracy	Scalability	Context	Reporting	Adaptability
Before (Manual Process)	Days to weeks to scan a single database	High risk of human error and inconsistency	Does not scale; infeasible for large data environments	Data fields are found without purpose/retention context	Ad-hoc, unstructured reports (e.g., spreadsheets)	Rules are informal and reside in people's heads
After (PriVista Solution)	Minutes to scan multiple datasets simultaneously	Consistent, rules-based detection with measurable precision/recall	Highly scalable; can be run on-demand via the web interface or scheduled automatically	Automatically links discovered fields to metadata, flagging gaps	Standardized, structured evidence inventory for audits and reviews	Configurable rules and data classes that can be easily updated

PriVista Software Architecture Layers



SPARKX

Data Flow Diagram (DFD)

The DFD below shows how data moves through PriVista — from user inputs to the final evidence inventory :

External Entities:

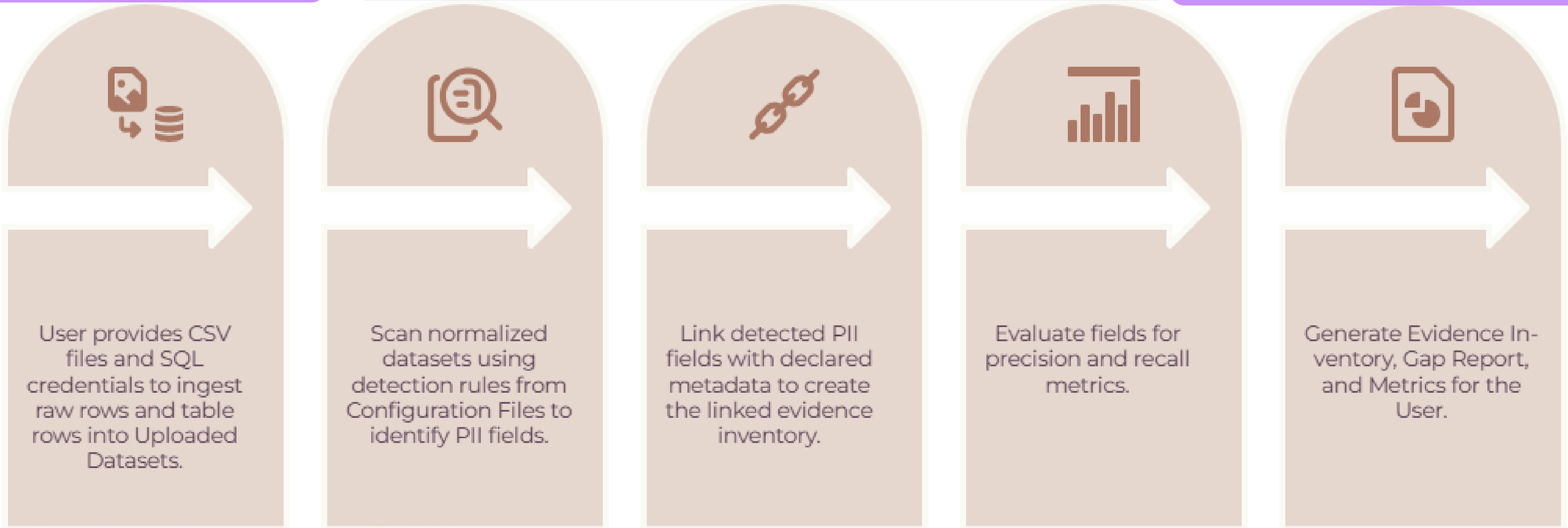
- **User** (DPO / Privacy Team)
- **Data Sources** (CSV files, SQL DB)

Processes:

1.0 Ingest Data → 2.0 Detect PII → 3.0 Link Metadata → 4.0 Evaluate → 5.0 Generate Report

Data Stores:

- **D1:** Uploaded datasets
- **D2**: Configuration (rules + metadata)
- **D3**: Evidence inventory



Deployment Timeline

Week 1: Foundation
Setup of private GitHub repository (with aiori-hackathon as collaborator). Detailed project planning and finalization of architecture.

Week 2: Core Development
Development of CSV and SQL connectors. Implementation of the basic Detection Engine with configurable regex rules.

Week 3: Feature Integration
Implementation of the Metadata Linking & Gap Analysis module. Creation of the Evidence Inventory Reporter.

Week 4: Evaluation & Demo
Development of the evaluation module for precision/recall. Final documentation, creation of the pseudocode snippet, and preparation of the final presentation/demo.

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Thank You for Your Attention!